

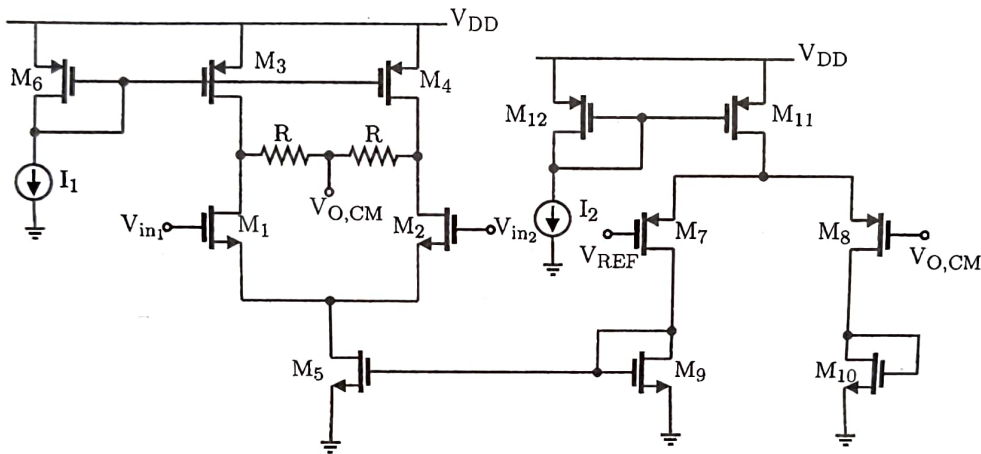
Instructions : Please put box around the final answers. Answers without legitimate units will be marked invalid. Overwritten answers will not be rechecked under any circumstances.

Average: 10.7 / 60

Question 1

For the following circuit, $I_1 = 50 \mu\text{A}$, $I_2 = 200 \mu\text{A}$, $\left(\frac{W}{L}\right) = \frac{50\mu\text{m}}{1\mu\text{m}}$ for all MOSFETS. $R = 10 \text{ M}\Omega$. What is the differential gain and common mode gain of main amplifier without common mode feedback? What is the optimum value of $V_{O,CM}$? With common mode feedback circuit, if common mode output voltage optimum value is allowed to vary by $\pm 1\%$, then what is the expected value of common mode gain? (write this value till 3 decimal places). What is CMRR without common mode feedback? What is common mode loop gain? What is common mode gain with feedback (achieved with circuit)? What is CMRR with common mode feedback (achieved with circuit)? [20]

Given: $V_{DD} = 1.8 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 50 \mu\text{A}/\text{V}^2$, $\lambda_p = 0.2 \text{ V}^{-1}$, $\lambda_n = 0.1 \text{ V}^{-1}$, $V_{th,n} = 0.4 \text{ V}$, $|V_{th,p}| = 0.5 \text{ V}$. Neglect body effect.

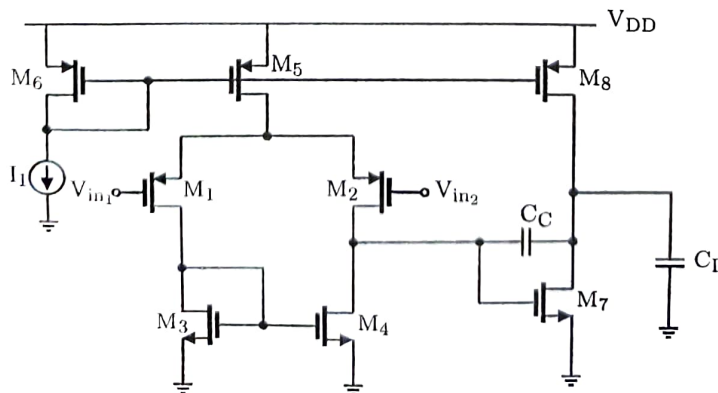


Question 2

For the following circuit, $I_1 = 10 \mu\text{A}$, $\left(\frac{W}{L}\right)_6 = \left(\frac{10\mu\text{m}}{1\mu\text{m}}\right)$, $\left(\frac{W}{L}\right)_5 = \left(\frac{20\mu\text{m}}{1\mu\text{m}}\right)$, $\left(\frac{W}{L}\right)_{1,2} = \left(\frac{10\mu\text{m}}{1\mu\text{m}}\right)$, $\left(\frac{W}{L}\right)_{3,4} = \left(\frac{10\mu\text{m}}{1\mu\text{m}}\right)$, $\left(\frac{W}{L}\right)_7 = \left(\frac{80\mu\text{m}}{1\mu\text{m}}\right)$, $C_C = 0.22 \text{ pF}$ and $C_L = 1 \text{ pF}$.

Calculate slew rate, Gain Bandwidth Product, DC gain, bandwidth, location of second pole, location of RHP zero and phase margin. [15]

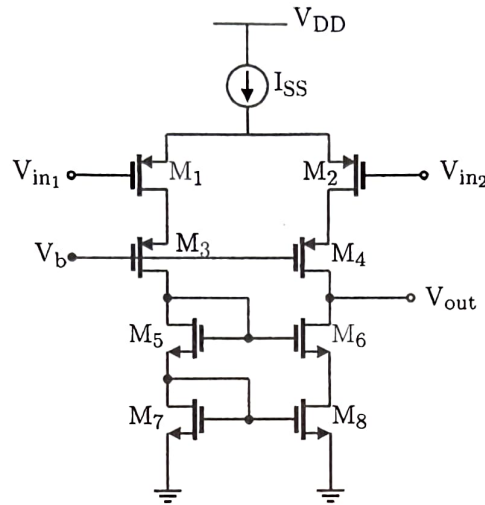
Given: $V_{DD} = 1.8 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 50 \mu\text{A}/\text{V}^2$, $\lambda_p = 0.2 \text{ V}^{-1}$, $\lambda_n = 0.1 \text{ V}^{-1}$, $V_{th0,n} = 0.4 \text{ V}$, $|V_{th0,p}| = 0.5 \text{ V}$.



Question 3

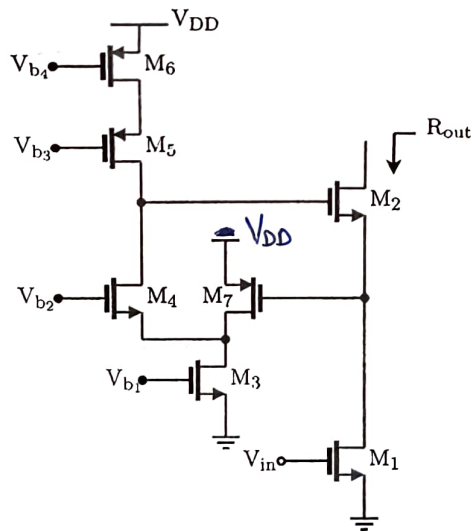
For the circuit given below, $I_{SS}=50\text{ }\mu\text{A}$, $V_{OV_{1-4}}=0.2\text{ V}$ and $V_{OV_{5-8}}=0.1\text{ V}$. Minimum voltage required across I_{SS} is 0.15 V , $V_b=0.7\text{ V}$. Calculate gain of the amplifier, $V_{out,max}$ (for given value of V_b), $V_{out,min}$. If V_{out} is shorted with V_{in2} , what is the value of $V_{out,max}$ and $V_{out,min}$? [10]

Given: $V_{DD}=1.8\text{ V}$, $\mu_n C_{ox}=100\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox}=50\text{ }\mu\text{A/V}^2$, $\lambda_p=0.2\text{ V}^{-1}$, $\lambda_n=0.1\text{ V}^{-1}$, $V_{th0,n}=0.4\text{ V}$, $|V_{th0,p}|=0.5\text{ V}$.

**Question 4**

In the figure given below, V_{OV} of all NMOS is 0.1 V and $|V_{OV}|$ of all PMOS is 0.2 V , $I_{D2}=I_{D4}=I_{D7}=10\text{ }\mu\text{A}$. What is the value of R_{out} ? What is the minimum and maximum value of V_{b2} ? [7]

Given: $V_{DD}=1.8\text{ V}$, $\mu_n C_{ox}=100\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox}=50\text{ }\mu\text{A/V}^2$, $\lambda_p=0.2\text{ V}^{-1}$, $\lambda_n=0.1\text{ V}^{-1}$, $V_{th0,n}=0.4\text{ V}$, $|V_{th0,p}|=0.5\text{ V}$.

**Question 5**

CMOS inverters 1 and 2 are symmetric. Consider the circuit below. (Technology parameters: $V_{DD}=5\text{ V}$, $V_{Tn}=|V_{Tp}|=0.8\text{ V}$, $k'_n=60\text{ }\mu\text{A/V}^2$, $k'_p=25\text{ }\mu\text{A/V}^2$, $L=1\text{ }\mu\text{m}$, ignore channel length modulation in transistors.)

If $V_{out1}=0.6\text{ V}$, determine V_{in1} and V_{out2} . [8]

